

7.3 Recommended Operating Conditions

Over-operating free-air temperature range (unless otherwise noted). See [§ 9.1.8](#) for more information on cell configurations. All voltages are relative to VSS, except "Cell input differential."

			MIN	TYP	MAX	UNIT
V _{BAT}	Supply voltage	(BAT – VSS)	6		25	V
		(BAT – VC5x), (VC5x – VSS)				
		(BAT – VC10x), (VC10x – VC5x), (VC5x – VSS)				
V _{IN}	Input voltage	Cell input pins, differential (VCn – VCn – 1) where n = 1..15/10/5 (BQ76940/BQ76930/BQ76920, respectively), in-use cells only	2		5	V
		(VCn – VSS) where n = 1..5	0		5 × n	V
		(VCn – VSS) where n = 1..5, (VCn – VC5x) where n = 6..10				
		(VCn – VSS) where n = 1..5, (VCn – VC5x) where n = 6..10, (VCn – VC10x) where n = 11..15				
		SRP	– 10		10	mV
		(VC0 – VSS)				
		(VC0 – VSS), (VC5b – VC5x)				
		(VC0 – VSS), (VC5b – VC5x), (VC10b – VC10x)				
		SRN	– 200		200	mV
		SCL, SDA				
		(TS1 – VSS)	0		3.6	V
		(TS1 – VSS), (TS2 – VC5x)				
		(TS1 – VSS), (TS2 – VC5x), (TS3 – VC10x)				
		REGSRC				
V _{OUT}	Output voltage	CHG, DSG	0		16	V
		REGOUT, ALERT	0		3.6	V
		(CAP1 – VSS)				
		(CAP1 – VSS), (CAP2 – VC5x)				
		(CAP1 – VSS), (CAP2 – VC5x), (CAP3 – VC10x)				
I _{CB}	Cell balancing current (internal, per cell)	BQ76920	0		50	mA
		BQ76930, BQ76940	0		5	mA
R _C	External cell input resistance	BQ76920	40	100	1K	Ω
		BQ76930, BQ76940	500	1K	1K	Ω
C _C	External cell input capacitance		0.1	1	10	μF
R _f	External supply filter resistance		40	100	1K	Ω
C _f	External supply filter capacitance		1	10	40	μF
R _{FILT}	Sense resistor filter resistance		100	1K		Ω
R _{ALERT}	ALERT pin to VSS resistor			1M		Ω
C _L	REGOUT loading capacitance		1	4.7		μF
C _{CAP}	REGSRC, CAP1, CAP2, and CAP3 output capacitance		1			μF
R _{TS}	External thermistor nominal resistance (103AT) at 25°C		10K			Ω